

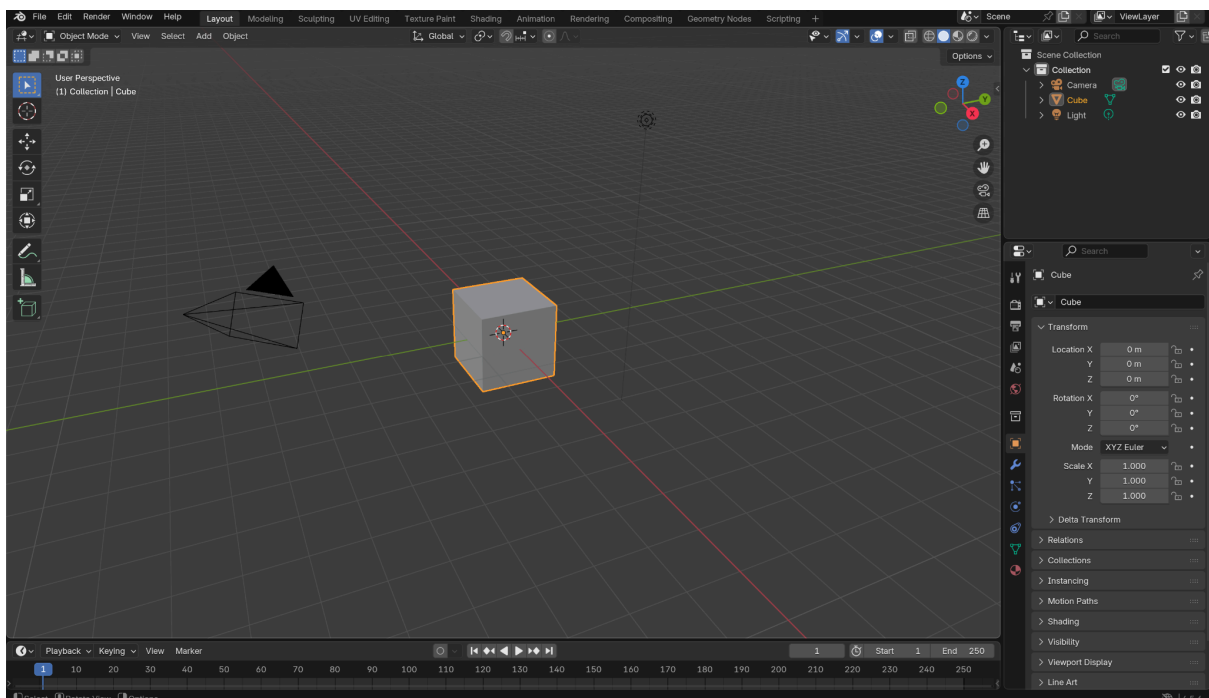
This intro to 3D modeling will guide you through the process of making a simple camera within blender, a free, open-source software.

If you ever feel lost on the controls, I've assorted the important ones used in this guide at the bottom of this document.



You'll open Blender to this screen (or one with a slightly different image, depending on the version.), Click 'General' on the left hand menu to create a new file.

Once gone, **press Ctrl + S** to save the file, name it and place it somewhere accessible.



Ta Da! Welcome to Blender!

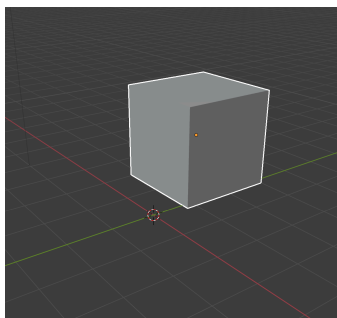
The top right hand side is the Outliner, this displays every item in the workspace including hidden items, lights, empties, etc. The bottom right-hand

side is the properties menu, generally used to alter properties of an item in the workspace.

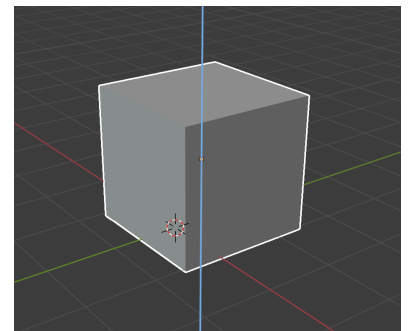
Hold the middle mouse button and move the mouse around, this is how you change the angle of approach. If you **hold shift while holding the middle mouse button** and move the mouse, you will move around the workspace.

Scroll the middle mouse button to zoom in and out, though there is a limit to how much you can do this (which only really matters if working on an incredibly small scale.)

In the centre of the screen you'll notice a cube - **left click** it to highlight it orange, you have now selected the cube. Alternatively, you can select the cube by clicking 'Cube' in the outliner menu. To focus on any object, **hold the backtick key** (located above the tab button, and left of '1') while having selected an item and hover your mouse over 'view selected', and then let go. This brings the centre of focus back to an object (which is especially useful for traversing!)



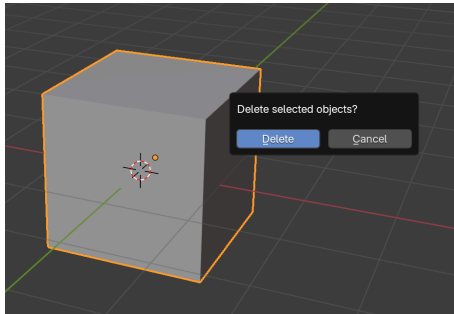
Select the cube again, and **press the G key**, this is how you move an object. If you also **press the middle mouse button while holding an object**, it will snap to a global axis. Keep pressing the middle mouse button and move the mouse to switch to a different axis.



Press the R key to rotate an object, **press the S key** to change the size of an object (proportionally), and **press the middle mouse button** while doing either of these actions to do it along a specific axis. **Left click** to confirm the change, or **right click** to cancel the change.

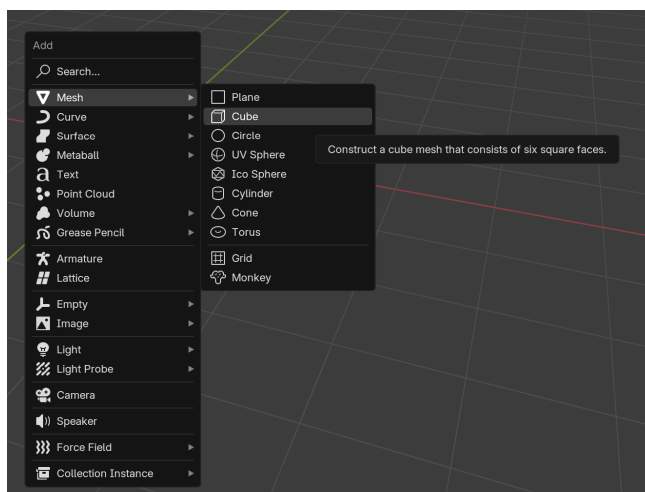
Press ctrl + Z to undo, or **shift + ctrl + Z** to redo.

For this intro, we will be making an incredibly simple camera - this is due to the fact it is made up of some rather simple shapes. Feel free to alter the look of your camera in any way you'd like, there is no 'right' way!



Begin by highlighting the cube we used in the intro, and **Press the X key** to delete it.

Press shift + A to open the insert menu, and add a cube.



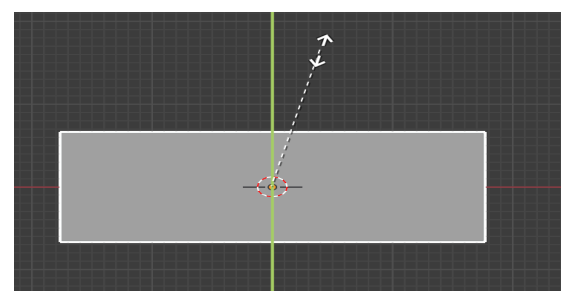
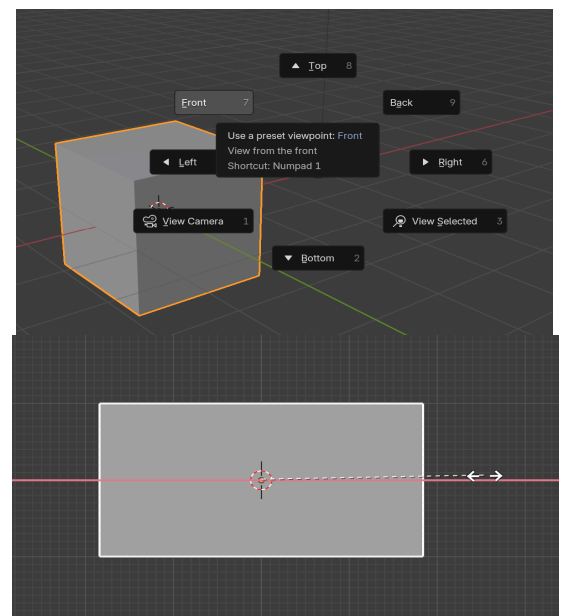
Why delete a cube and then add it again? First, to demonstrate how to delete an object, and second, there is a common habit held by people who use blender to do this exact thing for the sake of 'starting on a fresh slate'. Realistically it makes absolutely no difference, but it feels nice to do.

Now that you have a new cube, **hold the backtick button and hover over 'front'** then let go to be looking at the exact front of the cube.

Press the S key and hold the middle mouse button, then drag along the **red (X) axis** to lengthen the cube.

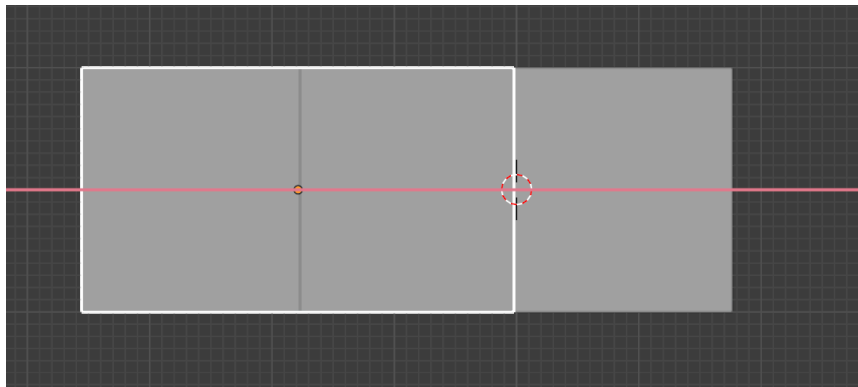
Hold the backtick button and hover over 'top' to be looking at the exact top of the cube. **Press the S key and hold the middle mouse button, then drag** along the **green (Y) axis** to make it thinner.

You will now have a rectangle shape. To look at it, you can **hold the middle mouse button** to look around it.

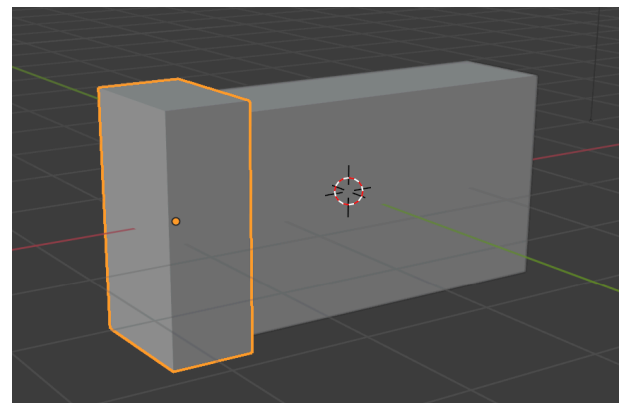


Look at the rectangle from the front again by **holding the backtick button and hover over 'front'**. Select it, and press Shift + D to duplicate it. *It is important to note that pressing the right mouse button will not cancel the duplication, so if you end up with too many of an object to get rid of it you will either have to undo (ctrl + z) or select the extra object by finding it in the outliner menu and deleting it by pressing x.*

Hold the middle mouse button while snapping to the **X axis**, and drag the copied rectangle to the edge of the first rectangle. **Left click** to confirm its position. Try to get the **orange dot** in the middle of the selected rectangle on the edge of the first rectangle, this is an easy way to have the centre of an object (signalled by the **orange dot** - the centre of mass) centred where you'd like it.



Next, change the shape of the rectangle by **pressing the S key and holding the middle mouse button**, switch to different views by **pressing the backtick key and hovering over the different options**. The goal here is to end up with handles for the camera, so make it look something like this.



We don't want to have to model another handle all over again, so what we'll do is add a modifier to this rectangle.

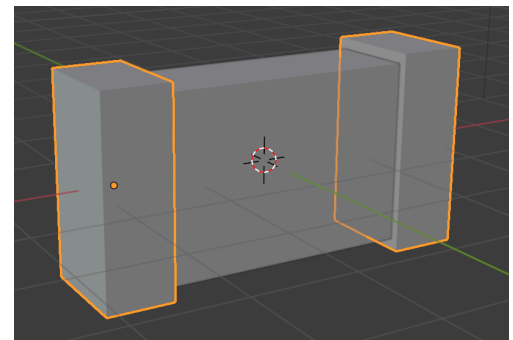
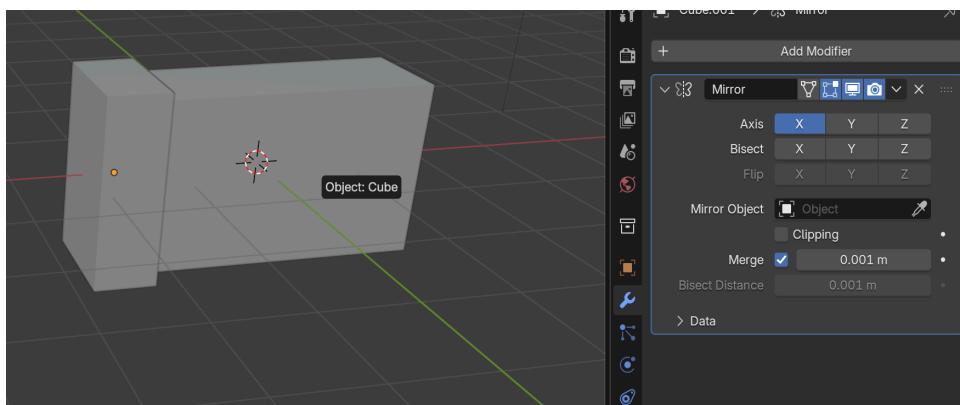
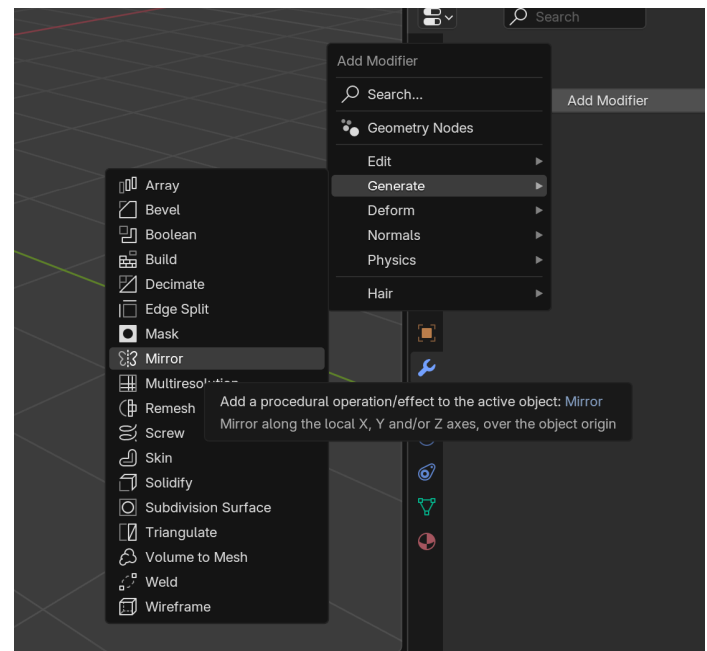
A modifier 'automatically alters an object's geometry or properties without permanently changing the underlying mesh'. This essentially means that it can deform or generate the model in some way with the option to remove the change

until it is confirmed. Common examples include 'mirror' which is what we are about to use, shrinkwrap, and bevel.

Look at the bottom right of your screen. When having selected the handle of the camera you'll notice a **blue wrench** in the option list, select that to get onto the modifier menu.

Press the 'add modifier' button, then locate 'Mirror' via Generate > Mirror. Press it to confirm.

Now that you have the mirror modifier selected, press the eyedropper icon in 'Mirror Object', and select the main body of the camera to mirror it.



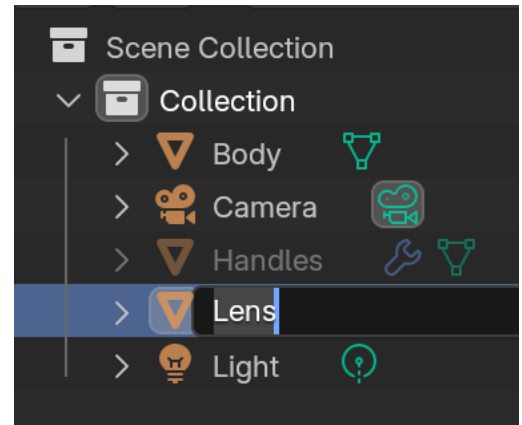
You should now have a mirrored handle! Since it is mirrored to the body of the camera, moving or rotating the body will also change the location of the mirrored handle, so for now we'll keep it where it is.

Next, we should make a lens for the camera. Select the handles again, and **press the H key**. This will temporarily hide them out of view, though they are still there! **Press alt + H** to unhide them if ever needed.

Select the cube, and look at it from the left view (*a reminder - if you ever forget the controls, you can look back through the instructions or scroll to the bottom of the document to find a collection of every control used in this tutorial.*) Press **shift + A** and a circle by finding it in Mesh > Circle. **Left click** to confirm.

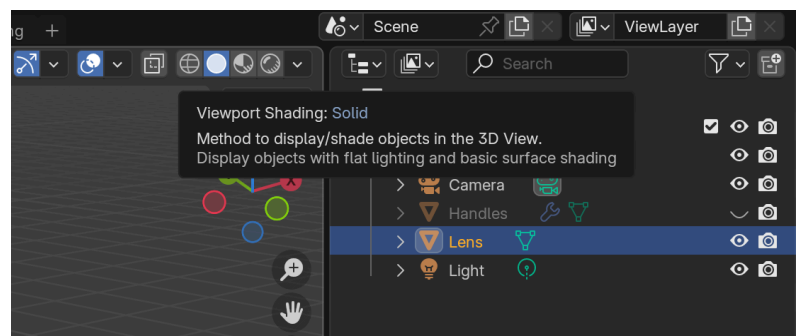
Now that we have three objects, this will be a good time to start naming them for ease of use. Look in the top right of the screen at the outliner menu, **double click** the body of the cube in this menu and rename it to 'Body', repeat this with the hidden handles to 'Handles' and the new circle to 'Lens'.

You'll notice a camera and light in this menu too, but we'll deal with those later.

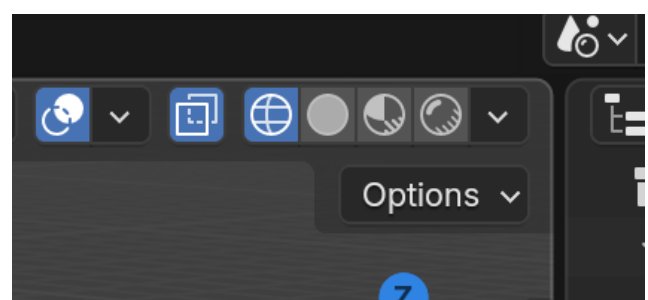


Go back to the left view of the camera and select the Lens. **Press the R key and type (yes, type into your keyboard) 90. Left click (or press enter) to confirm.** This rotates the object along the **red (X) axis**, and by typing 90 you are rotating the object exactly 90 degrees along that axis, this can be done with any number.

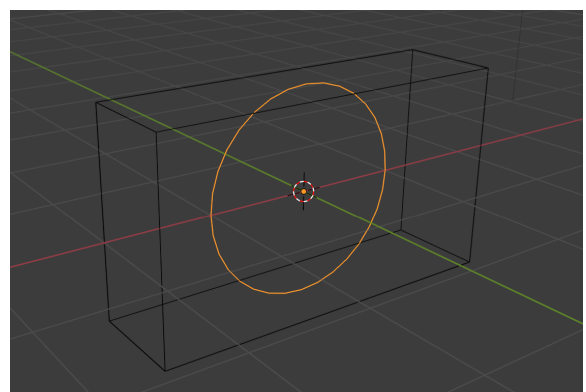
Now, we'll introduce views! Look at the top right of the screen, you'll notice 4 buttons with circles in them, the second one will be selected. These are *views*. The one we are currently on is called the viewport solid view, essentially an unrendered, uncoloured view of the model - it is the easiest to make a model in.

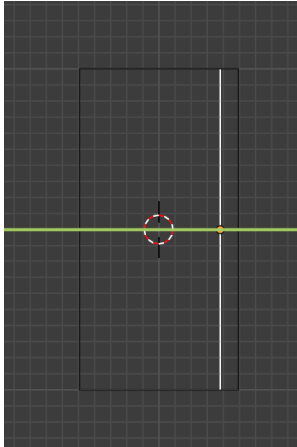


We want to look at the wireframe of our camera, so select the first circle in the options.



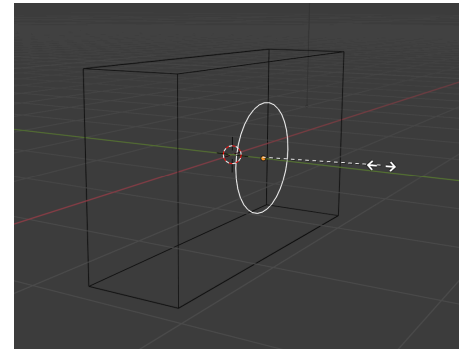
This will show you the wireframe of your model - where every edge exists. We will go into this more in depth (alongside the other views) later, for now, we are using it because it is see through.





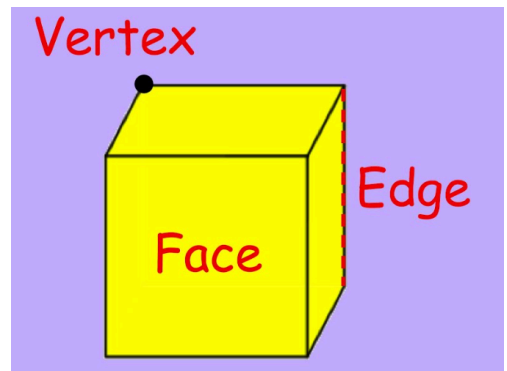
Look at the body of the camera from the left again, select the lens, and **press G** to move it, then **press the middle mouse button** to move it along the **green (Y) axis** to the front of the camera, just before the edge.

Press S and move the mouse to make the lens smaller. **Left click** to confirm



Before we continue with the camera, we should quickly learn about what a mesh is. By definition, it is 'a set of points and cells connected to form a network'. In terms of 3D, this refers to Vertices, edges, and faces (triangles).

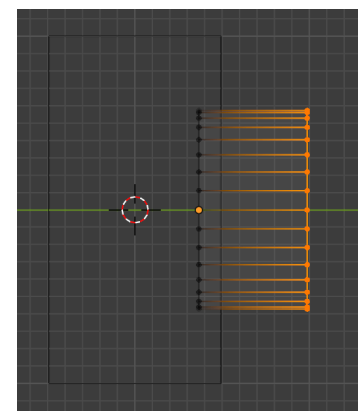
Every model is made up of these components, and that is what a mesh is, a collection of them that are connected in some way.

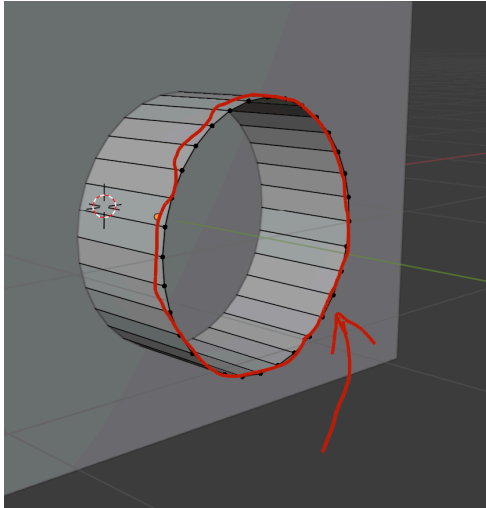


Within real-life use, a smaller 'vert and tris count' (vertex and face count) generally means greater performance as it is less for a computer of some sort to have to process and render, and this is why you'll often hear the terms 'optimization' and 'topology' in 3D modeling, they are very closely intertwined.

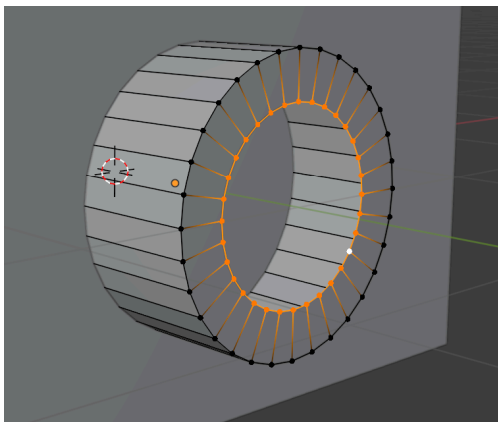
(If you have any questions about this - feel free to ask me! It's one of my favourite topics in 3D modeling, and makes up for a lot of what goes on in terms of game development.)

Select the lens, and look at it from the left view. **Press the Tab key** to open edit mode. This shows you the vertices, edges, and faces of a mesh. **Press A** to select every vertice, and then **Press E** to extrude the vertices. Drag these right along the **green (Y) axis**, and then press the left mouse button to confirm.

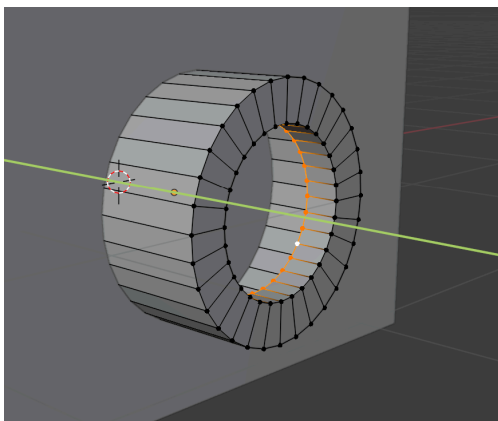




Go back into the Solid viewport view (second one along), and look at your camera from a front angled view (Go into the front view of your camera, and hold the middle mouse button to move the angle so it's more so diagonally looking at the camera). Select the lens again, and you'll notice that you can see a circle of vertices at the lens of the camera. **Deselect any vertices by pressing anywhere on the screen without a vertex.** Hover over an edge of this circle, and **hold the alt key while left clicking.** This will select what is called an 'edge loop'.



Press E to extrude, and then **press S** to size the newly extruded vertices. Move the mouse to make them smaller than the current circle.

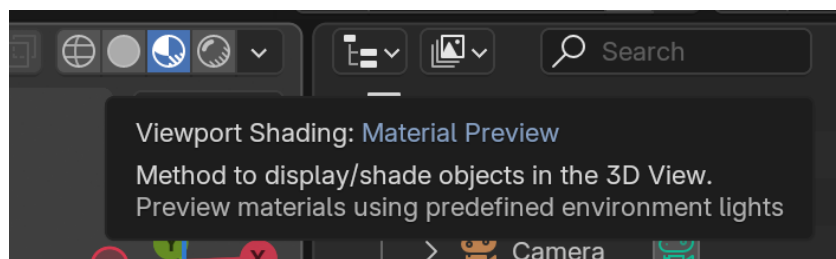


Press E again to extrude, this time **pressing the middle mouse button** to snap to the **green (Y) axis**, and bring those vertices inside of the model.

Finally, while still having that new inner edge loop selected, **press F** to fill. This will fill that space with a face.

Press Tab to exit edit mode, and **press alt + H** to unhide the handles. Move the camera around by **holding the middle mouse button** to look around your model.

For the sake of keeping things simple for now, we will move onto colour. On the view menu, click the third one along - Material Preview.



Begin by selecting the body of the camera. Look at the bottom right of the screen at the properties menu, and select the very bottom red option - Material. **Press 'new'.**

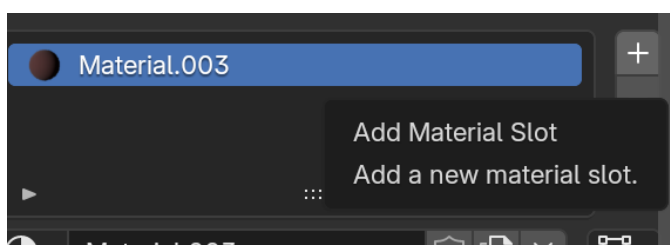
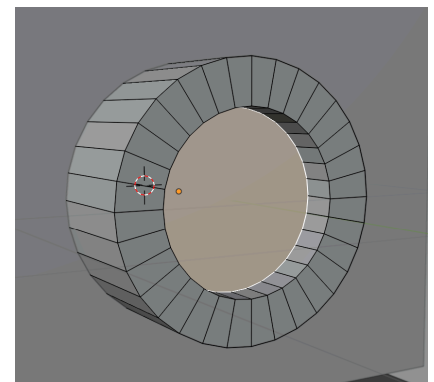
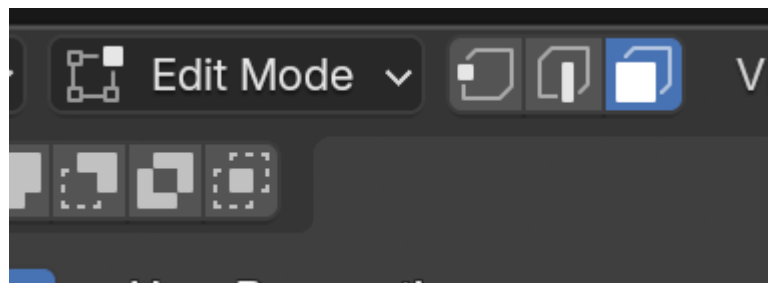
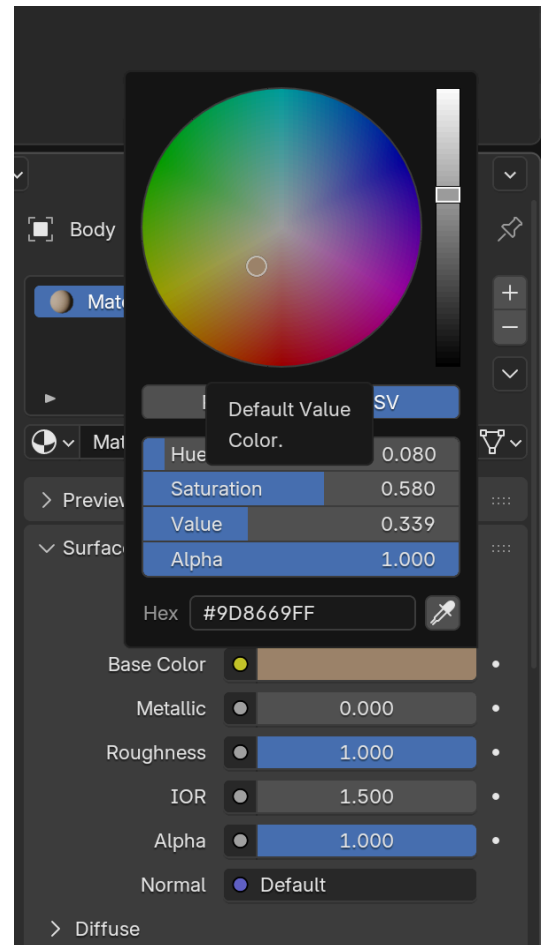
The automatic shader Blender uses is called 'Principled BSDF', it is essentially an all in one shader that is very versatile.

Feel free to mess around with these values, for now I have turned roughness to 1 - which makes the model matte rather than shiny. Change the colour to your liking. Repeat this process with the handles and lens to your liking.

You'll notice the 'lens' of your lens model will have the same colour as the rest. We want this to be shiny for our purposes.

Select the lens mode, and **press Tab** to go into edit mode. Look at the top left of your screen, you'll notice right of where it says 'edit mode' there are cubes with 3 different options. These allow you to select between editing vertices, edges, or faces. Select the third option for faces.

Select the lens face we made, then look back at the material properties. We want to add a material slot, so click the + button, and then press assign with the face selected. This allows one mesh to have several different materials without having to separate the model into separate pieces



Now you should have a coloured Model! *If you'd like a more detailed effect shading wise, at the top of your screen you will notice a 'shading' tab. Select that, and you will find that you are able to move blocks of nodes around. **Press shift + A** to add more. I'd suggest searching up tutorials if you're looking for a specific style.*

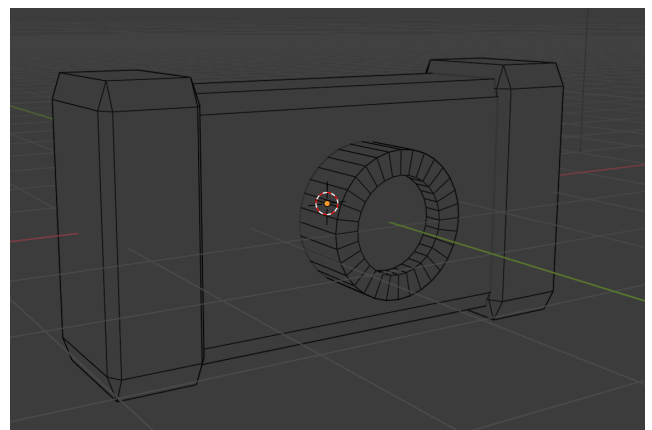
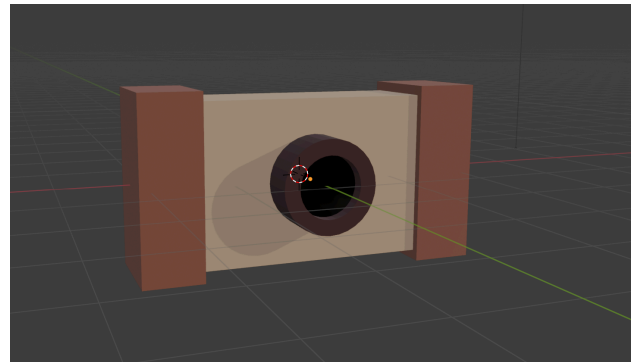
Now... we have a camera, but it looks rather plain, doesn't it?

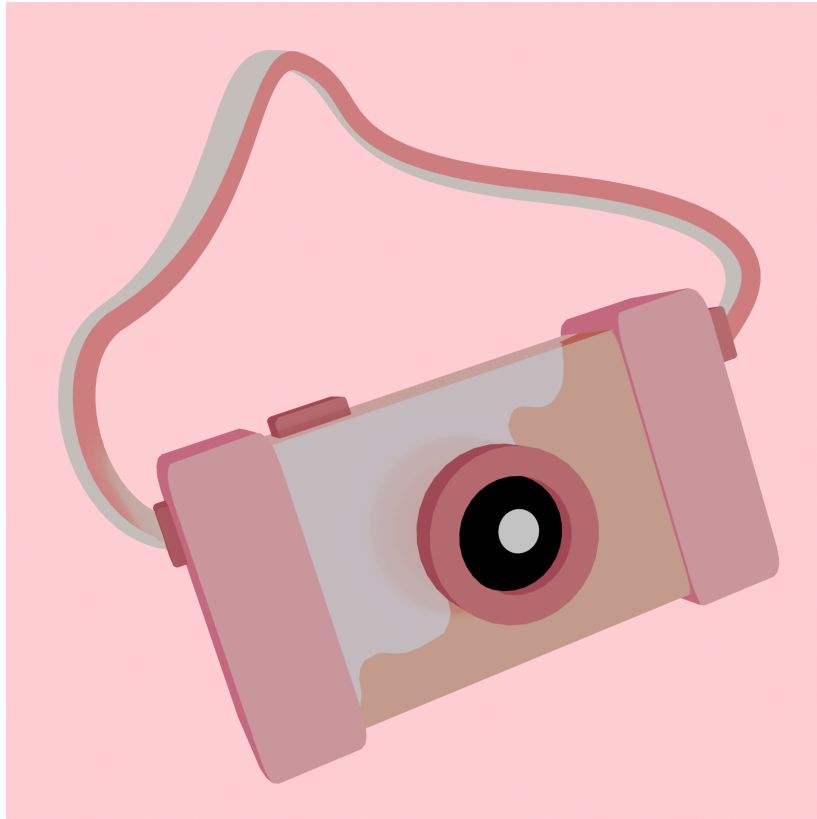
A simple way to make a plain model more interesting is to use bevel. To do this, select the body of the camera and go into edit mode, then **press A** to select all vertices (ensuring you are back in vertice select mode!)

Having selected every vertex, **Press ctrl + B** and move the mouse, this will bevel every selected edge - meaning one edge becomes two, smoothing it out. **Press the left mouse button** to confirm. Repeat this process with the handles.

Beveled edges are an incredibly useful tool, and if you **Press ctrl + B and scroll the middle mouse button**, you can increase and decrease the amount of new loop cuts that are made.

Feel free to add any more details you'd like to your camera, in a few minutes I added a strap, button, and added a simple wave design to the front. I also made what's known as a 'Toon shader' - where instead of having many colours for lights, darks, and mids of a shadow it's split into only a few, in my case 3.





Once you have knowledge of the controls, Blender is an incredibly versatile tool. If you'd like to continue to learn Blender i'd suggest the 'Donut Tutorial' which is the most popular on youtube, it goes over every facet of blender in a simplified manner.

Here is a collection of the most important controls used in this tutorial

Press G while having selected an object

Moves an object

Left mouse button

Select an object

Hold middle mouse button and move mouse

Moves camera from a focus point

Hold shift and move mouse while holding middle mouse button

Moves camera

Press shift + A

Add mesh or nodes

Press A

Select all

Press H

Hide whatever is selected

Press H + alt

Unhide All

Press Tab

Enter and exit edit mode

Scroll middle mouse button

Zoom in and out

Hold Middle Mouse button

Snap to axis